FACULTY OF ENGINEERING AND TECHNOLOGY

BACHELOR OF TECHNOLOGY

COMPETITIVE CODING

(303105259)

4TH SEMESTER

(2ND YEAR)

Artificial Intelligence & Machine

Learning Department

PARUL UNIVERSITY

FACULTY OF ENGINEERING & TECHNOLOGY

PARUL INSTITUTE OF ENGINEERING & TECHNOLOGY

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

STUDENT DETAILS

NAME	:	
ENROLLMENT NO	:	
SUBJECT	:	COMPETITIVE CODING
SUBJECT CODE	:	303105259
SEMESTER	:	4TH
FACULTY NAME	:	

Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY

BACHELOR OF TECHNOLOGY

CERTIFICATE

This is to certify that Mr./Ms_____with Enrollment no._____Has successfully completed his/her Laboratory Experiments in the COMPETITIVE CODING (303105259) from the Department of Artificial Intelligence & Machine Learning **during the academic year 2025-2026**.

Date of Submission: .....

Staff Incharge:..... **Head of Department:.....**

Faculty of Engineering & Technology
Subject Name: Competitive Coding
Subject Code: 303105259
B.Tech. CSE Year 2 Semester 4

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Sr. No.	Experiment Title	Page No.		Date		Mark	Sign
		From	To	Performance	Assessment		
1.	Write a program for implementing a MINSTACK which should support operations like push, pop, overflow, underflow, display 1. Construct a stack of N-capacity 2. Push elements 3. Pop elements 4. Top element 5. Retrieve the min element from the stack						
2.	Write a program to deal with real- world situations where Stack data structure is widely used Evaluation of expression: Stacks are used to evaluate expressions, especially in languages that use postfix or prefix notation. Operators and operands are pushed onto the stack, and operations are performed based on the LIFO principle.						
3.	Write a program for finding NGE NEXT GREATER ELEMENT from an array.						
4.	Write a program to design a circular queue(k) which Should implement the below functions a. Enqueue b. Dequeue c. Front d. Rear						
5.	Write a Program for an infix expression and convert it to postfix notation. Use a queue to implement the Shunting Yard Algorithm for expression conversion.						
6.	Write a Program for finding the Product of the three largest Distinct Elements. Use a Priority Queue to efficiently find and remove the largest elements.						
7.	Write a Program to Merge two linked lists(sorted)						
8.	Write a Program to find the Merge point of two linked lists(sorted).						
9.	Write a Program to Swap Nodes pairwise.						
10.	Write a Program for Building a Function ISVALID to VALIDATE BST.						
11.	Write a Program to Build BST						

12.	Write a Program to determine the depth of a given Tree by Implementing MAXDEPTH.						
13.	Write a Program to Understand and implement Tree traversals i.e. Pre-Order Post-Order, In-Order						
14.	Write a Program to perform Boundary Traversal on BST.						
15.	Write a program for Lowest Common Ancestors						
16.	Write a Program to verify and validate mirrored trees or not.						
17.	Write a Program for a basic hash function in a programming language of your choice. Demonstrate its usage to store and retrieve key-value pairs						
18.	Implement a hash table using separate chaining for collision handling. Perform operations like insertion, deletion, and search on the hash table.						
19.	Write a Program to Implement Two sums using HASHMAP.						
20.	Write a Program to Implement Search, insert, and Remove in Trie.						
21.	Write a Program to Implement Huffman coding.						
22.	Write a Program to find Distinct substrings in a string.						
23.	Write a Program to find The No of Words in a Trie.						
24.	Write a Program to view a tree from left View.						
25.	Write a Program to Traverse a Tree using Level Order Traversal.						